

a Physical causal world

Typed state · process laws · instruments · private parameters

b Experimental interaction runtime

Validate · commit or reject · measure · account · replay

c Task and evaluation contract

Public goal · legal operations · budget · termination · metrics

External researcher

LLM agent
Classical optimizer
Human researcher



d Frozen platform qualification

64

Task-world units

1,786

Boundary / categorical recipes

192

Invalid-action probes

52

Generated compositions

Generated cases: 18 new topologies + 8 new identities + 26 coverage cases

Additional checks: 32 module probes · 7 valid interfaces · 7 invalid compositions

Finite-domain execution consistency; separate from 240 autonomous campaigns and wet-lab validity.